Air Monitoring and Management System

Catalogue

airqoon.com

V5.2-EN - Preliminary version

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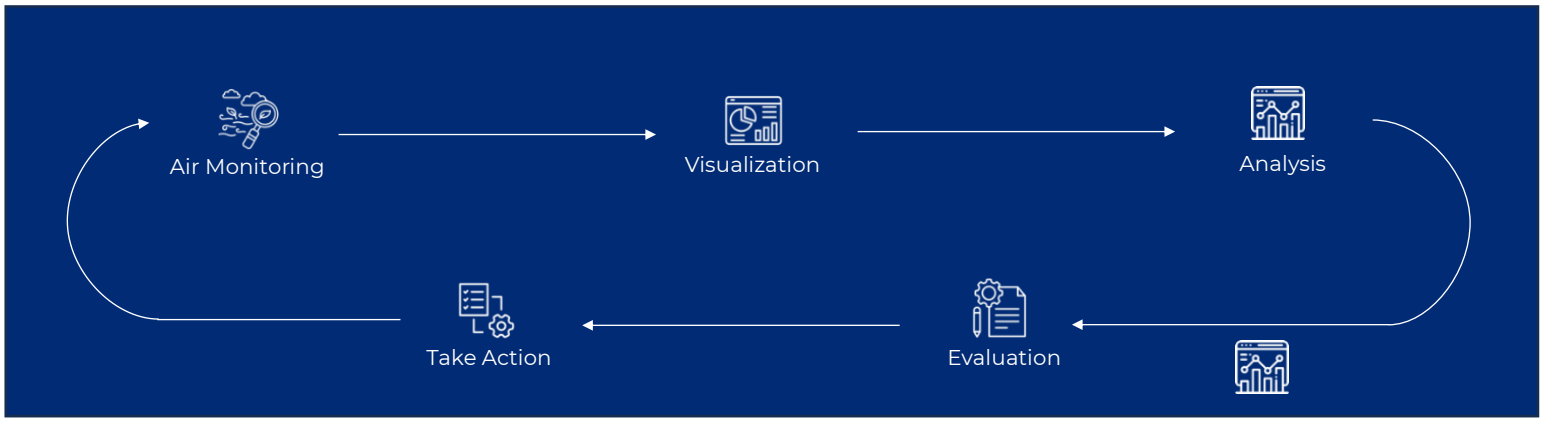
About Us

Reliable and easy-to-use air monitoring at scale

Airqoon plays an important role by providing low-cost, reliable, and user-friendly solutions to air pollution. Airqoon's main goal from day one is to leverage technology to provide accessible and scalable solutions to the public and private sectors to help them understand the air around them and find policies/solutions around it.

Founded in 2018, airqoon has extensive experience in understanding and raising awareness of risks to the environment and human health through sensor technology and data science. Our mission is to contribute to creating a clean environment and a healthy living space for future generations.



For Local Authorities

Access to real-time air quality data enables local governments to understand air quality, prioritize resources more effectively, and make data-driven decisions.

For Businesses

Real-time air quality monitoring keeps industrial plants moving harmoniously and responsibly while helping to protect the environment and human health.

Airqoon Sensor Unit Series Unit L

The Airqoon Sensor Unit L is a cost-effective and easy-to-use sensor system that measures real-time ambient air particulate matter and gaseous pollutants in the air.

Ease of maintenance

Unit L can be easily deployed and become operating in less than 5 minutes. Diagnostic information is collected in the airqoon cloud.

Accurate and reliable data

Airqoon unit's measurement accuracy is increased significantly with the airqoon's smart calibration process

Seamless data collection and analysis

Unit L works from the start together with the airqoon cloud

Global connectivity

Unit L provides continuous and seamless monitoring with global connectivity. It can work with 4G and NB-IoT.

Energy self-sufficient

Its onboard battery pack and solar panel allow airqoon units to work 24/7/365 without interference.

KEY BENEFITS

CONTINUOUS MONITORING

Unit L offers real-time monitoring of air quality parameters and this allows for immediate detection of changes or spikes in pollution levels, enabling prompt actions to be taken.

LOCALIZED DATA

Unit L provides data at a local level, allowing for monitoring specific areas of interest such as neighborhoods, schools, workplaces, or even personal environments.

DATA-DRIVEN DECISION MAKING

Policymakers use the environmental data to develop targeted strategies, implement pollution control measures, and evaluate the effectiveness of existing environmental regulations.

Parameter	Specification
Sampling Period	1, 5, 10, 15, 30, 60 minutes
Unit Life Expectancy	+5 years*
Connectivity	Cellular 4G/NB-IoT, WiFi
Operating Conditions	-10°C to +65°C and 10% to 99% RH
Mounting	Kits available for pole and wall mounting
Size	35cm x 25cm x 18cm
Weight	3.2kg with solar panel
Enclosure	IP34
Powering Modes	Solar panel or 5V DC power source
Battery	3.7V 15.2Ah
Sensor Lifetime	Electrochemical: 2 years*, PID-based: 5 years (excluding replaceable lamp and electronic stack), NDIR-based: +5 years*, Nephelometric: +5 years*
Compliance	CE

*(depending on the preffered sampling period and environmental conditions)

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KEY BENEFITS

AWARENESS AND EDUCATION

Airqoon air pollution and management system plays a crucial role in raising awareness about air pollution and its impacts on human health. By providing accessible and easy-to-understand data, sensors empower individuals, communities, and policymakers to make informed decisions and take preventive measures.

RESEARCH AND INNOVATION

Airqoon air pollution and management system contributes to ongoing research and innovation in the field of air quality monitoring. The data collected from Airqoon Sensor Unit L can be used to improve pollution modeling, develop new technologies, and advance scientific understanding of air pollution and its sources.

Sensor Performance

The airqoon air pollution monitoring and management system offers diverse product packaging options tailored to specific applications and use cases.

Sensor Name	Sensor Type	Accuracy	Range
PM2.5	Laser scattering (nephelometric)	r2* >0.9	0 to 1000 µg/m3 w/ 1µg/m3 resolution
PM10	Laser scattering (nephelometric)	r2* >0.9	0 to 1000 µg/m3 w/ 1µg/m3 resolution
NO2	Electrochemical	r2* > 0.82, LODs**: 2 ppb	0 to 5000 ppb w/ 1 ppb resolution
O3	Electrochemical	r2* > 0.82, LOD**: 5 ppb	0 to 5000 ppb w/ 1 ppb resolution
SO2	Electrochemical	r2* > 0.7, LOD**: 5 ppb	0 to 5000 ppb w/ 1 ppb resolution
CO	Electrochemical	r2* > 0.8, LOD**: 10 ppb	0 to 10 ppm w/ 10 ppb resolution
Temperature	Integrated env. sensor	0°C-65°C ±1.0°C 0°C-20°C ±1.25°C	-20°C to 60°C
Humidity	Integrated env. sensor	20-80%RH at 25°C ±3%RH	0% to 100%
Pressure	Integrated env. sensor	At 0°C-65°C ±1 hPa	300Pa to 1100hPa

PM10 in µg/m3 is calculated from PM2.5 readings and calibrated with respect to a FEM instrument.
*r2-scores are calculated during co-location trials with respect to a FEM instrument.
**LOD: Limit of Detection

Packaging Options

Packaging	Included Components
Lite	PM2.5, PM10, Ambient Temperature, Relative Humidity and Pressure
Gas-1	Lite + 1 regular Gas
Gas-2	Lite + 2 regular Gas
Gas-3	Lite + 3 regular Gas
Gas-4	Lite + 4 regular Gas
Gas-Special	*Keep in touch for more details*
External Addons	Noise Sensor (Class 2 IEC 61672-1:2013) Wind Speed / Direction (Anemometer), Solar Panel, 5V Adapter, Extended Solar Power Package (ESP)

Optional gaseous pollutants:

- **Regular**: NO, NO2, O3, SO2, CO
- **Special**: CO2, H2S, VOC, CH4, N2O

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Unit M

Airqoon Sensor Unit M is designed to monitor air quality in indoor public spaces, providing a cost-effective and portable sensor system.

Smooth Data Analysis

Unit M operates seamlessly with airqoon cloud from the outset; providing the capability to visualize, analyze, and manage your data in the cloud.

Accurate and reliable data

Airqoon unit's measurement accuracy is increased significantly with the airqoon's smart calibration process

Ease of maintenance

Airqoon units can be easily deployed and up and running in less than 5 minutes.

Global connectivity

Unit M provides continuous and seamless monitoring with global connectivity. It can work with 4G and NB-IoT.

Specific Pollutants

Unit M provides the capability to simultaneously measure PM1, PM4, PM2.5, PM10, CO2, and VOC pollutants.

KEY BENEFITS

CONTINUOUS MONITORING AND ANALYSIS

Unit M provides remote monitoring and data collection capabilities. Contiunuous monitoring allows for the identification of pollution trends and patterns. Airqoon can easily analyze all data and generate reports for your decision-making process.

SECURE CONNECTIONS AND DATA TRANSMISSION

Airqoon enterprise data platform is designed to prevent unauthorized access, data maniupulation, and security breaches through encrypted data transmission over secure connections. You can also share collected data via API or export it in various formats and seamlessly integrate data from third-party devices.

Parameter	Specification
Sampling Period	1, 5, 10, 15, 30, 60 minutes
Unit Life Expectancy	+10 years
Connectivity	WiFi, BLE, 4G and NB-IoT
Operating Conditions	-20°C to +60°C and 10% to 99% RH
Mounting	Kits available for pole and wall mounting
Size	12cm x 12cm x 6cm
Weight	<1kg
Enclosure	IP34
Powering Requirements	5V DC power source
Battery	3.7V 7.6 Ah
Sensor Lifetime	Nephelometric: +5 years*
Compliance	CE

*(depending on the preffered sampling period and environmental conditions)

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Sensor Performance

The airqoon air pollution monitoring and management system offers diverse product packaging options tailored to specific applications and use cases.

Sensor Name	Sensor Type	Accuracy	Range
PM1	Laser scattering (nephelometric)	0-100 µg/m3 ±5µg/m3 >100 µg/m3 ±10%	0 to 1000 µg/m3 w/ 1µg/m3 resolution
PM2.5	Laser scattering (nephelometric)	0-100 µg/m3 ±5µg/m3 >100 µg/m3 ±10%	0 to 1000 µg/m3 w/ 1µg/m3 resolution
PM4	Laser scattering (nephelometric)	0-100 µg/m3 ±25µg/m3 >100 µg/m3 ±25%	0 to 1000 µg/m3 w/ 1µg/m3 resolution
PM10	Laser scattering (nephelometric)	0-100 µg/m3 ±25µg/m3 >100 µg/m3 ±25%	0 to 1000 µg/m3 w/ 1µg/m3 resolution
CO2	NDIR	±40 ppm +5 % (400ppm-5000ppm)	400 to 5000 ppm
tVOC	MOx type	-	0.3 to 30 ppm Ethanol in clean air
Temperature	Integrated env. sensor	0°C-65°C ±1.0°C 0°C-20°C ±1.25°C	-20°C to 60°C
Humidity	Integrated env. sensor	30-70%RH at 25°C ±3%RH	0% to 100%

KEY BENEFITS

FLEXIBLE AND CUSTOMIZABLE USE

You can esaily integrate and configure each sensor, including start time, sampling times and contaminant alarm limits to meet individual product requirements.

LEGAL COMPLIANCE AND TRACEABILITY

Airqoon's sensor unit significantly streamlines continuous monitoring processes and ensures compliance with regulatory requirements. Additionally, it facilitates the efficient generation of reports to uphold quality standards and meet compliance requirements based on the acquired data.

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Unit N

Airqoon Sensor Unit N is a network-based sensor system developed for the accurate, continuous, and reliable measurement of environmental noise in urban and industrial areas. The sensor does not record sounds but monitors changes in ambient noise levels.

KEY BENEFITS

REAL-TIME MONITORING

Enables immediate intervention in environmental noise pollution by tracking noise levels in real-time.

REMOTE ACCESS

Data from monitoring devices are transmitted to a central system via remote communication methods. This allows centralized management of noise sources in different locations.

COST-EFFECTIVENESS

Offers large-scale data collection and analysis at a lower cost compared to traditional noise measurement methods, enabling noise monitoring in more areas and providing a budget-friendly solution.

Compact Design

Offers a lightweight and portable structure, adaptable to ant environment, providing a flexible solution that is easy to position and resistant to outdoor conditions.

Certified Calibration

Provides a measurement system calibrated by an ISO 17025 accredited laboratory with international traceability.

Accurate Measurement

Delivers high-accuracy environmental noise measurements in compliance with IEC 61672 Type 2 standards.

Global connectivity

Unit N provides continuous and seamless monitoring with global connectivity. It can work with 4G and NB-IoT.

Smooth Data Analysis

Unit N operates seamlessly with airqoon cloud from the outset; providing the capability to visualize, analyze, and manage your data in the cloud.

Parameter	Specification
Range	30-130 dB
Accuracy	±3dB@23±5℃,accordance with IEC 61672 standard type 2, calibrated at 94dB(1kHz) input
Frequency Response	31.5Hz - 8kHz.
Corrector	B&K 4226
Microphone	Capacitive microphone , size: 0.5 inch
Supply	5VDC, 12-24VDC
Output	RS485
Ingress Protection	IP65
Power Consumption	<20mW
Response Time	<200ms
Operating Temperature	-10℃-+50℃@5-80%RH
Storage	-40-70℃@20%-90%RH
Shell Material	ABS & 304SS

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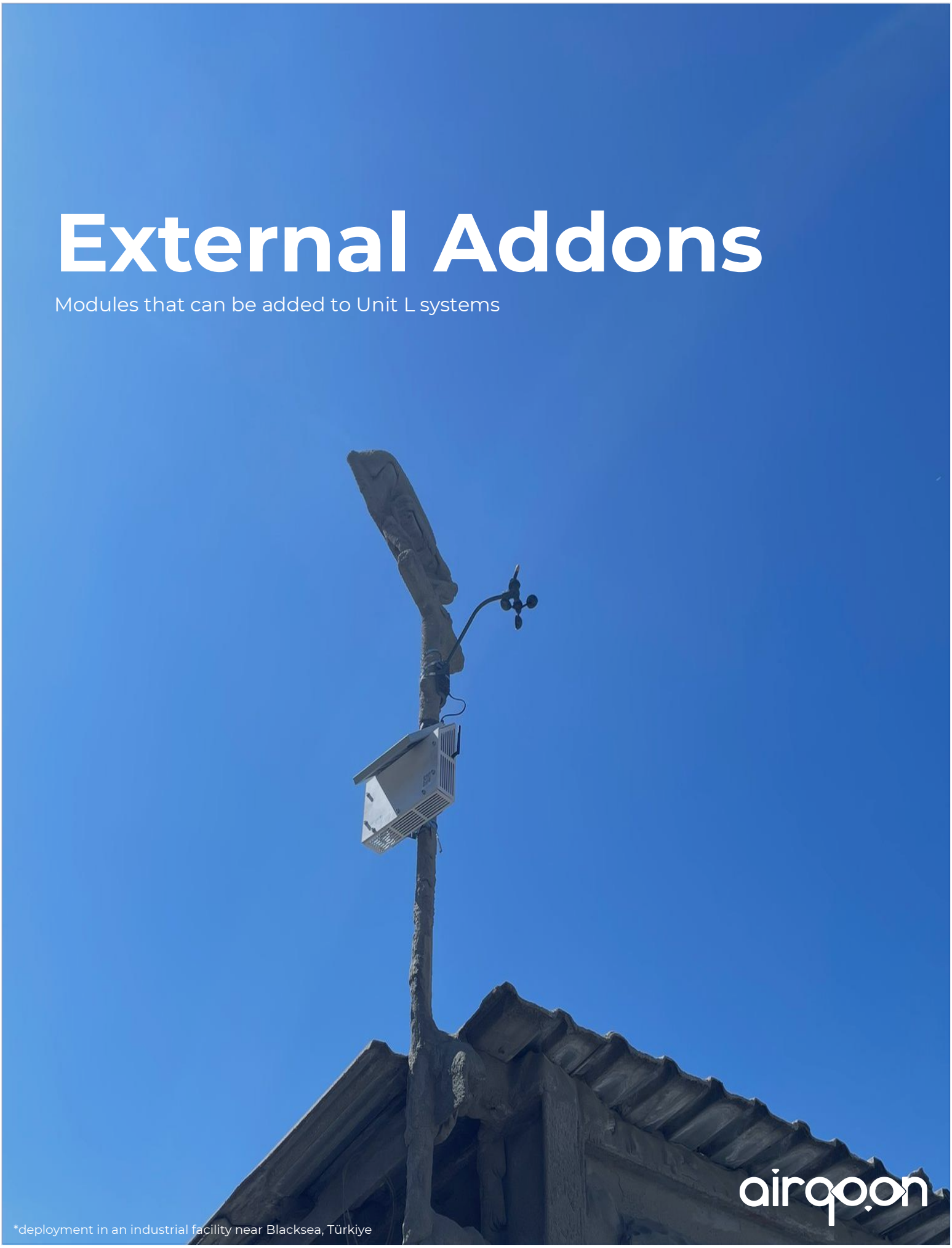
External Addons

Modules that can be added to Unit L systems



*deployment in an industrial facility near Blacksea, Türkiye

Noise Sensor

The airqoon air pollution monitoring and management system offers diverse product packaging options tailored to specific applications and use cases.

Parameter	Specification
Range	30-130dB
Accuracy	±3dB@23±5°C, accordance with IEC 61672 standard type 2*, calibrated at 94dB (1 kHz) input
Frequency Response	31.5Hz – 8kHz.
Microphone	Capacitive microphone, size: 0.5 inch
Response Time	<200ms
Operating Temperature	-10°C -+50°C@5-80%RH

*(Calibration certificate from an accredited lab (ISO 17025) is provided upon request)

Features

- High Sensitivity
- Low Consumption
- Fast Response
- Long Service Life

Anemometer - Wind Speed/Direction

The anemometer measures wind speed and direction, which is critical for assessing airflow patterns and their impact on air quality and polutant dispersion.

Parameter	Specification
Wind Speed Range	1-200 mph / 0.5-89 m/s
Test Speed	200 mph+
Bearings	Stainless steel
Weight	1.5 kg

Features

- High Accuarcy
- Waterproof
- 360° Wind Direction
- Wireless Data

Extended Solar Power Package (ESP)

The ESP provides enhanced solar power solutions with larger panels and external battery packs, ensuring continuous operatin in low-sunlight environments.

Parameter	Specification
Extended Solar Panel Options	12W, 25W, 50W
External Battery Capacity	3.7V, 15.2Ah
Power Output	5V, 2A
Interface	Through airqoon units power input socket

Features

- Multipanel options
- Plug-and-play
- Additional battery pack
- Continuous power

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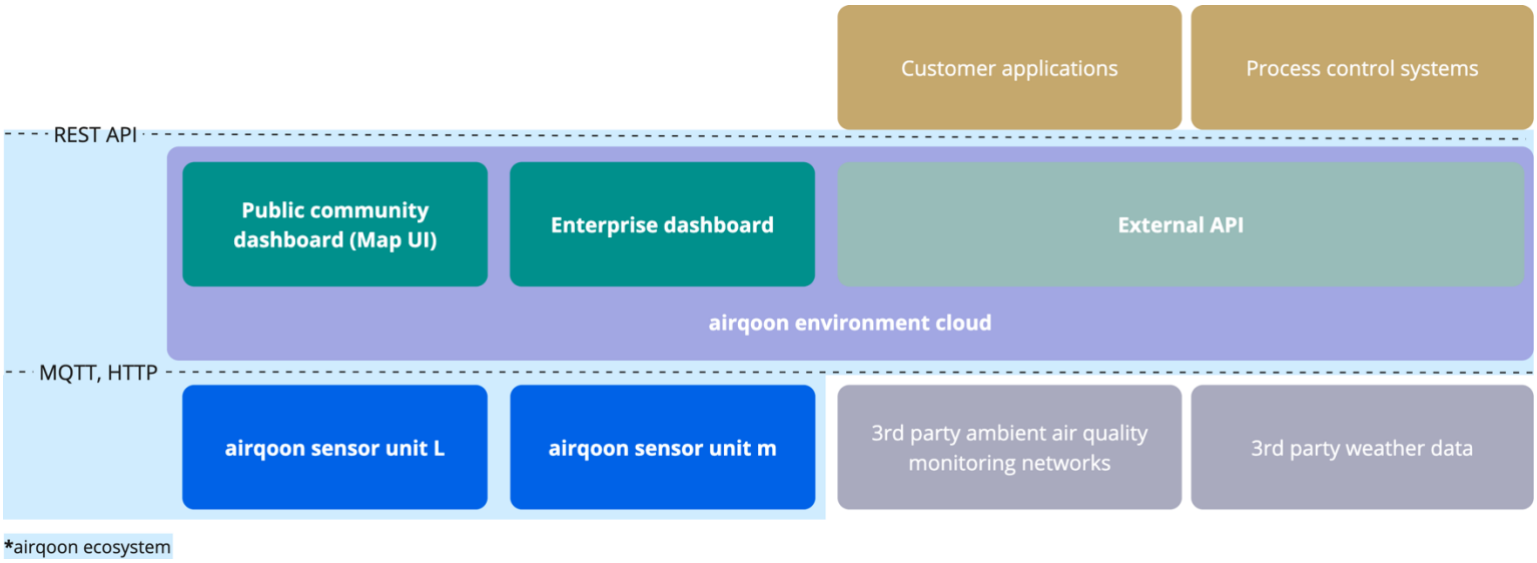
Solution Architecture



Airqoon Environment Cloud

Airqoon air pollution monitoring and management system is a cloud-based software used to manage, store and analyze air quality data collected by airqoon sensor units. Sensor units collect air quality data by measuring various pollutants in the atmosphere. This data is transmitted to the airqoon cloud after being detected by the sensors.

Airqoon Environment Cloud receives this data and stores it securely, providing capabilities to visualize and analyze air quality data. The data can be presented through graphs, tables, and maps via a user-friendly interface. These features enable easy monitoring of the current air quality status and historical trends.

Our AI-powered analysis tools enhance data analysis by identifying key trends and automatically highlighting them in reports. This feature particularly supports decision-making processes by providing valuable insights in large datasets.

Public: Air quality data collected in the airqoon cloud can be shared with the public through the public dashboards. This map shows the air quality index and pollution levels in a specific area on the map. Thus, this information helps the public monitor the current state of air quality and make informed decisions.

Enterprise: Air quality data can be pretty critical if it is tied to a manufacturing process of an enterprise. In terms of security, the airqoon cloud keeps the collected data confidential. Airqoon Environment Cloud is also offered with a user management feature to enterprises that would like to manage the air around their industrial facility. Data is stored securely and protected against unauthorized access. Users are given the ability to monitor, manage and react quickly to air quality data.

External API: The external API will provide a set of predefined methods and protocols that external developers can use to integrate airqoon's air quality data and analysis capabilities into their own applications, systems, or services. Beyond the functions provided by airqoon's user interface, this could open up a range of possibilities for using air quality data in a variety of ways.

Use Cases

*After the disastrous eartquakes in 6th February 2023, Kahramanmaras, Türkiye

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Air Quality Network

Airqoon units are deployed throughout a city to monitor air quality in real time3. They measure parameters such as particulate matter (PM2.5, PM10), nitrogen dioxide (NO2), and ozone (O3). By collecting data on air pollution levels, cities can identify pollution hotspots, assess the effectiveness of pollution mitigation measures, and make informed decisions to improve air quality. The transparent and accessible presentation of data with a public dashboard encourages public awareness and action.

Access to accurate and real-time air quality data helps local governments and communities gain the ability to effectively prioritize and allocate resources by identifying the most critical air quality issues.

Benefits from airqoon

- Compatible with a wide range of parameters including PM, gases, and meteorological parameters
- Tools to share real-time air quality information with citizens and communities
- A complete solution to measure, analyze, and monitor real-time air quality.
- Essential tool to point out the sources of pollution and plan corrective actions

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Enviromental Compliance

Industries need to comply with environmental regulations and monitor emissions. Monitoring industrial emissions helps identify processes that produce the most gases and particulates, thereby reducing your environmental impact and improving worker health. Low-cost continuous air monitoring sensors can detect and measure pollutants such as volatile organic compounds (VOCs), carbon monoxide (CO), sulfur dioxide (SO2), and nitrogen oxides (NOx).

Airqoon provides real-time monitoring, assisting industries in ensuring compliance and taking corrective actions if necessary, while also helping businesses adopt cleaner practices and measure changes by assessing the impact of operations on air quality.

Benefits from airqoon

Alerts when thresholds are exceeded. Necessary tool in determining the negative impact of which actions or processes

Generates a secure and encrypted password combination for user login. Data is securely stored and protected against unauthorized access

Real-time monitoring of multi-pollutants to detect leaks and emissions

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Impact Assessment of Green Technologies

In an era where environmental sustainability is gaining increasing importance, measuring and monitoring the environmental impact of green investments is crucial to promoting environmentally conscious and sustainable business practices. In this context, investments directed towards green technology can bring about significant environmental benefits, such as reducing fossil fuels, promoting clean energy sources, and adopting eco-friendly practices.

Airqoon provides real-time, data-driven, actionable environmental intelligence by accurately assessing pollution levels in production facilities. Thus, it enables the measurement of the environmental impact of investments, understanding how much change they can bring, and establishes a solid foundation for progressing closer to environmental sustainability goals.

Benefits from airqoon

Robust and compact design designed for harsh environments suitable for all research work

Reliable and accurate data of different parameters in real-time

Provides analytics that enrich environmental reports and aid in organized decision making

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Air Quality Control

Children, due to their ongoing developmental processes, inhale more air per unit of body weight compared to adults, making them more sensitive and vulnerable to the effects of pollutants. For healthy growth and development, children require safe and clean play areas. hopping mall playgrounds are spaces where children spend a significant amount of time, making the safety and hygiene of these areas critically important for their overall health and well-being.

To ensure a safe play environment for children in shopping mall playgrounds, airqoon continuously monitor indoor air quality, observing ventilation systems, temperature, humidity, and various pollutant levels. This enables us to swiftly and effectively intervene in case of potential pollution situations.

Benefits from airqoon

Increased social responsibility and the possibility of continuously monitoring the air quality of certain pollutant trends

Automatically receive alerts or alarm signals when certain air quality parameters are exceeded

The air quality monitoring system can be integrated into existing building management systems or smart building systems

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Case Studies

Inegol/Bursa Municipality Air Quality Monitoring Network

. Sensor unit L, airqoon enviroment cloud

In cooperation with the Municipality of Inegol (Bursa), we have installed airqoon units in and around the D100 highway area, where there is intense industrial activity and 60 thousand vehicles pass daily.

Kadikoy/Istanbul Municipality Environmental Monitoring and Early Warning System

. Sensor unit L, airqoon enviroment cloud

Kadikoy (Istanbul) Municipality has developed cooperation with airqoon to track air pollution and heat waves affected by climate change and to develop an early warning system within the scope of the Increasing Social Resilience in the Climate Adaptation Process Project, which was entitled to receive a grant from the European Union in 2020-2021.

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Avcilar/Istanbul Municipality Air Quality Monitoring Network

. Sensor unit L, airqoon enviroment cloud

We have collaborated with the municipality to address air pollution in Avcilar (Istanbul), given its location on a major transportation axis, proximity to industrial areas, and maritime traffic.

Ambient Air Quality Monitoring at the Eastern Anatolia Observatory Erzurum, Turkiye

. Sensor unit L, airqoon enviroment cloud

The collaboration with airqoon aims to generate and interpret more comprehensive data for research on the impact of natural pollutants and the relationship between climate change and air pollution.

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Neomarin Children’s Playgrounds Air Quality Monitoring

. Sensor unit M, airqoon enviroment cloud

At Neomarin Play Area (Istanbul), which welcomes children between the ages of 0-14, we continuously monitor indoor air quality thanks to airqoon sensor unit M.

Air Quality Monitoring in the Playground KidZania Entertainment Center

. Sensor unit M, airqoon enviroment cloud

In collaboration with KidZania (Istanbul), we measure particulate matter, VOC, CO2, temperature, and humidity values in children's play areas to ensure their health and well-being.

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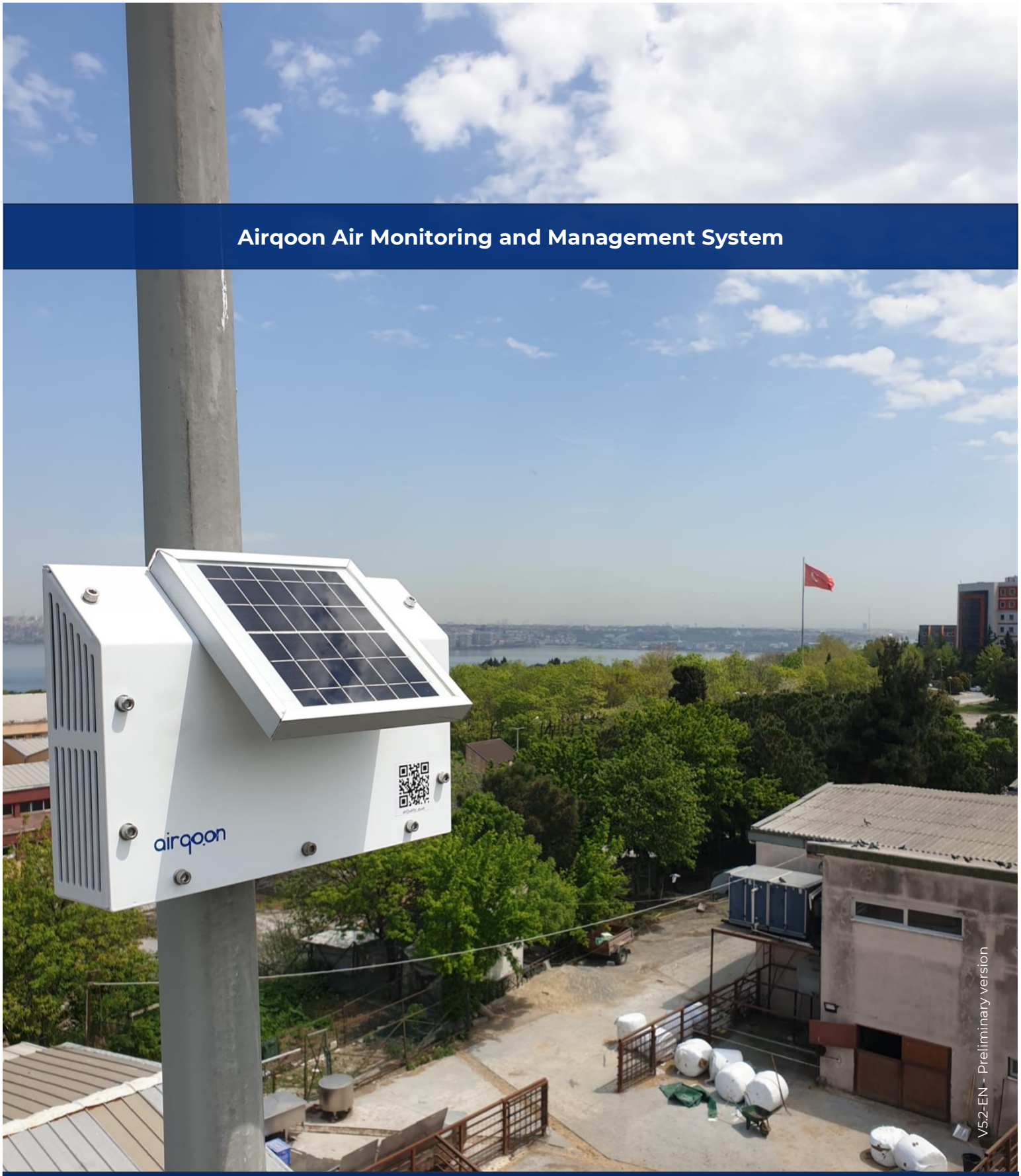
Airqoon Air Monitoring and Management System



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